

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Cause & Effect Matrix- BBOU, ELEP, AKAB
Document Number	EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-009
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Cause & Effect Matrix- BBOU, ELEP, AKAB

R01	28.06.22	Issued for Review	P.Kiran	P.Roy	W.Irem	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

R01	28.06.22	Issued for Review	P.Kiran	P.Roy			
Rev No	Date	Reason for Issue / Change	Prepared	Reviewed	TCE Approval		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02)

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-009

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES	
Document No.	Description

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EFFECT CAUSE		CAUSE AND EFFECT (BBOU)																							
		Indication/Alarm/Shutdown at SCADA																							
		Flow Indication	Pressure Indication/Alarm	Temperature Indication	Pressure Indication/Alarm	Pressure Indication/Alarm	Pressure Indication/Alarm	Close Inlet SDV (Shutdown)	Open Flare BDV (Blowdown)	Close Outlet SDV(Shutdown)	Temperature Indication	Temperature Indication	Temperature Indication	Pressure at CBD Vessel	Temperature Indication	Level Indication	Level Indication	Gas Release High Alarm	Gas Release High High Alarm	Gas Release High Alarm	Gas Release High High Alarm	Gas Release High Alarm	Gas Release High High Alarm	Gas Release High Alarm	Gas Release High High Alarm
DEVICE ID	SERVICE	BBOU-FI-1001	BBOU-PIPAL/PAH-1001	BBOU-TI-1001	BBOU-PIPAL/PAH-1002	BBOU-PIPAL/PAH-1003	BBOU-PIPAL/PAH-1004	BBOU-SDV-1001	BBOU-BDV-1002	BBOU-SDV-1003	BBOU-PIPAL/PAH-1013	BBOU-PIPAL/PAH-1014	BBOU-PIPAL/PAH-1015	BBOU-PI-1016	BBOU-TI-1003	BBOU-LI/LA/LAH-1001	BBOU-LI/LA/LAH-1002	BBOU-AAH-1001	BBOU-AAHH-1001	BBOU-AAH-1002	BBOU-AAHH-1002	BBOU-AAH-1003	BBOU-AAHH-1003	BBOU-AAH-1004	BBOU-AAHH-1004
BBOU-DPT-1001	Check Flow 1 -Flow Measurement	●																							●
BBOU-DPT-1002	Check Flow 2- Flow Measurement	●																							●
BBOU-PIT-1001	Battery Limit Pressure		●																						●
BBOU-TIT-1001	Battery Limit Temperature			●																					●
BBOU-PIT-1002	Flow Line Pressure				●																				●
BBOU-PIT-1003	Flow Line Pressure					●																			●
BBOU-PIT-1004	Flow Line Pressure						●		●	●															●
	2oo3 of BBOU-PIT-1002/1003/1004							●	●	●															●
BBOU-PIT-1013	Flow Line Outlet Pressure										●														●
BBOU-PIT-1014	Flow Line Outlet Pressure											●													●
BBOU-PIT-1015	Flow Line Outlet Pressure								●	●			●												●
	2oo3 of BBOU-PIT-1013/1014/1015							●	●	●															●
BBOU-PIT-1016	Pressure at Closed Drain Vessel													●											●
BBOU-TIT-1003	Temperature at Closed Drain Vessel														●										●
BBOU-LIT-1001	Level at Closed Drain Vessel															●									●
BBOU-LIT-1002	Level at Slop Tank								●	●	●						●								●
BBOU-SD-1001A	Remote Well Shutdown							●	●	●															●
BBOU-SPDC-XXXX	Signal From SPDC																								●
BBOU-AIT-1001	Point Type HC Gas Detector																●	●							●
BBOU-AIT-1002	Point Type HC Gas Detector																	●	●	●					●
BBOU-AIT-1003	Open Path HC Gas Detector																		●	●					●
BBOU-AIT-1004	Open Path HC Gas Detector																				●	●			●
BBOU-MAC-1001A	Manual Alarm Call Point 1																								●
BBOU-MAC-1002A	Manual Alarm Call Point 2																								●
BBOU-MAC-1003A	Manual Alarm Call Point 3																								●
BBOU-MAC-1004A	Manual Alarm Call Point 4																								●
BBOU-PL-1001	Pipeline Leak Detection																								●
																									●
																									●
																									●
																									●

CAUSE EFFECT			Indicator/Alarm/Shutdown at SCADA																																							
			ELEP-FI-1001	ELEP-PI/PAU/PAH-1001	ELEP-TI-1001	ELEP-PI/PAU/PAH-1002	ELEP-PI/PAU/PAH-1003	ELEP-PI/PAU/PAH-1004	ELEP-SDV-1001	ELEP-BDV-1002	ELEP-SDV-1003	ELEP-PI/PAU/PAH-1013	ELEP-PI/PAU/PAH-1014	ELEP-PI/PAU/PAH-1015	ELEP-PI/PAU/PAH-1010	ELEP-TI-1003	ELEP-PI/PAU/PAH-1018	ELEP-PI/PAU/PAH-1019	ELEP-PI/PAU/PAH-1020	ELEP-SDV-1004	BBOU-PI-1021	ELEP-TI-1004	ELEP-LIU/LUL/AH-1001	ELEP-LIU/LUL/AH-1002	ELEP-AAH-1001	ELEP-AAHH-1001	ELEP-AAH-1002	ELEP-AAHH-1002	ELEP-AAH-1003	ELEP-AAHH-1003	ELEP-AAH-1004	ELEP-AAHH-1004	ELEP-AAH-1005	ELEP-AAHH-1005	ELEP-AAH-1006	ELEP-AAHH-1006	ELEP-SPDC-XXXX	ELEP-PLA-1001	All Signals	ELEP-HB-1004		
DEVICE ID	SERVICE	Flow Indication	Pressure Indication/Alarm	Temperature Indication	Pressure Indication/Alarm	Pressure Indication/Alarm	Pressure Indication/Alarm	Close Inlet SDV (Shutdown)	Open Flare BDV (Blowdown)	Close Outlet SDV(Shutdown)	Temperature Indication	Temperature Indication	Temperature Indication	Pressure Indication/Alarm	Temperature Indication	Pressure Indication/Alarm	Pressure Indication/Alarm	Pressure Indication/Alarm	Close Inlet SDV (Shutdown)	Pressure at CBD Vessel	Temperature Indication	Level Indication	Level Indication	Gas Release High Alarm	Gas Release High High Alarm	Gas Release High Alarm	Gas Release High High Alarm	Gas Release High Alarm	Gas Release High High Alarm	Gas Release High Alarm	Gas Release High High Alarm	Gas Release High Alarm	Gas Release High High Alarm	SPDC Alarm Signal	Pipeline Leak Detection	Signal at Remote DCS	Hooper and Beacon					
ELEP-DPT-1001	Check Flow 1 -Flow Measurement	●																																				●				
ELEP-DPT-1002	Check Flow 2- Flow Measurement	●																																				●				
ELEP-PIT-1001	Battery Limit Pressure		●																																			●				
ELEP-TIT-1001	Battery Limit Temperature			●																																		●				
ELEP-PIT-1002	Flow Line Pressure				●																																	●				
ELEP-PIT-1003	Flow Line Pressure					●																																●				
ELEP-PIT-1004	Flow Line Pressure						●																															●				
	2oo3 of ELEP-PIT-1002/1003/1004							●	●	●																												●				
ELEP-PIT-1013	Flow Line Outlet Pressure										●																											●				
ELEP-PIT-1014	Flow Line Outlet Pressure											●																										●				
ELEP-PIT-1015	Flow Line Outlet Pressure												●																									●				
	2oo3 of ELEP-PIT-1013/1014/1015							●	●	●																												●				
BBOU-PIT-1021	Pressure at Closed Drain Vessel																			●																		●				
ELEP-TIT-1004	Temperature at Closed Drain Vessel																				●																	●				
ELEP-LIT-1001	Level at Closed Drain Vessel																					●																●				
ELEP-LIT-1002	Level at Stop Tank																						●															●				
ELEP-SD-1001A	Remote Well Shutdown							●	●	●																												●				
ELEP-SPDC-XXXX	Signal From SPDC																																					●				
ELEP-AIT-1001	Point Type HC Gas Detector																							●	●														●			
ELEP-AIT-1002	Point Type HC Gas Detector																								●		●												●			
ELEP-AIT-1003	Open Path HC Gas Detector																									●													●			
ELEP-AIT-1004	Open Path HC Gas Detector																										●												●			
ELEP-AIT-1005	Point Type HC Gas Detector																											●											●			
ELEP-AIT-1006	Open Path HC Gas Detector																												●										●			
ELEP-MAC-1001A	Manual Alarm Call Point 1																																						●			
ELEP-MAC-1002A	Manual Alarm Call Point 2																																						●			
ELEP-MAC-1003A	Manual Alarm Call Point 3																																						●			
ELEP-MAC-1004A	Manual Alarm Call Point 4																																						●			
ELEP-MAC-1005A	Manual Alarm Call Point 5																																						●			
ELEP-PIT-1010	Battery Limit Press. at the Inlet of Pig Receiver ELEP-A-200													●																									●			
ELEP-TIT-1003	Battery Limit Temp. at the Inlet of Pig Receiver ELEP-A-200														●																							●				
ELEP-PIT-1018	Flowline Pressure for interlock-ELEP-A-300 Launcher															●																						●				
ELEP-PIT-1019	Flowline Pressure for interlock-ELEP-A-300 Launcher																●																					●				
ELEP-PIT-1020	Flowline Pressure for interlock-ELEP-A-300 Launcher																	●																				●				
	2oo3 of ELEP-PIT-1019/1020/1021							●												●																		●				
ELEP-PL-1001	Pipeline Leak Detection																																				●	●				

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